

Autonomous Systems and Robotics 700

Intelligent Autonomous Robotic Systems Project

*Component A: Perception, Motion Planning and Control for an Autonomous
Mobile Robot*

*Component B: Multi-Agent Robotic Swarm System with Learning-Based
Decision-Making*

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Written report

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1. Introduction

This report presents the design, implementation and evaluation of two intelligent robotic systems: (A) a mobile robot for perception, shortest-path planning and closed-loop control, and (B) a cooperative swarm using consensus, particle swarm optimisation (PSO) and a learned navigation policy. The systems use Webots and Python libraries, with Matplotlib for required visualisations; ROS and Gazebo are not used.

Both systems use the same 10×10 m arena defined in `common/env.py`, keeping the simulation layout consistent across experiments. Section 2 describes the methodology, Section 3 presents quantitative results, Section 4 discusses trade-offs and ethical issues, and Section 5 concludes the project.

2. Methodology

2.1 Component A: Perception, Planning and Control

Sensor simulation and perception. A 360° LiDAR was ray-cast from 16 survey poses with Gaussian range noise ($\sigma = 0.02$ m) and 1% dropout. A synthetic top-down camera was processed with OpenCV using smoothing, HSV segmentation, morphology and contours. LiDAR and camera evidence were fused in a log-odds occupancy grid; camera likelihood ratios were calibrated on held-out renders.

Motion planning. The fused occupancy grid was inflated by the robot radius plus a tested safety margin. Dijkstra and A* were implemented on an 8-connected grid with corner-cutting disabled. Euclidean, octile and Manhattan heuristics were compared, followed by line-of-sight path simplification and fixed-spacing resampling. The A* method is based on the heuristic search approach of Hart, Nilsson and Raphael (2021).

Control system design. The robot uses the unicycle model $\dot{x} = v \cos\theta$, $\dot{y} = v \sin\theta$, $\dot{\theta} = \omega$, with first-order actuator lags ($\tau_v = 0.20$ s, $\tau_\omega = 0.15$ s) and a 0.10 s transport delay. A PID controller with derivative-on-measurement, integral separation and output saturation was tuned by ITAE grid search under a 0.1 rad/s angular-drift disturbance (Hart, Nilsson and Raphael, 2021).

System integration. Perception, planning and control operate in one closed loop. Pure-pursuit selects a path target, PID controls heading error, speed is reduced when heading is poorly aligned, and a front-LiDAR safety layer slows the robot near unexpected obstacles. Pose feedback includes Gaussian noise and constant angular drift (Hart, Nilsson and Raphael, 2021).

2.2 Component B: Multi-Agent Swarm with Learning-Based Decision-Making

Simulation setup. E-puck robots were modelled in the Python/Matplotlib simulator and mirrored in Webots using a generated world from the same arena definition. Webots controllers reuse the same PID gains and trained network weights, allowing the simulation to reproduce the tested algorithms. Consensus and swarm intelligence. Average consensus was evaluated on ring, line, range-limited and complete communication topologies, including random link failures. Max-ID flooding was used for leader election. PSO performed distributed task allocation using global-best (gbest) and local-best ring (lbest) communication; results were compared with exact enumeration, greedy allocation and random assignment. PSO follows Kennedy and Eberhart (1995), while consensus is based on the networked multi-agent framework of Olfati-Saber, Fax and Murray (2024).

Machine learning. A two-hidden-layer MLP (32 ReLU units per layer) was trained by imitation learning to map a 16-ray LiDAR sweep and target bearing (sin/cos) to three steering actions. Data were split 70/15/15 for training, validation and testing. Performance was measured using test accuracy, a confusion matrix, learning curves and 200 closed-loop navigation episodes (Olfati-Saber, Fax and Murray, 2024).

Centralised versus distributed coordination. A coverage task visited a 0.5 m grid until 99% of reachable free space was covered. The centralised architecture used one coordinator with global knowledge; the distributed architecture merged local and neighbour information and claimed targets without a coordinator. Both were tested over 10 seeds under nominal conditions, robot failure, coordinator failure, reduced communication range and team sizes from 1–8 (Olfati-Saber, Fax and Murray, 2024).

3. Results and Evaluation

3.1 Component A Results

3.1.1 Perception and sensor fusion

Table 1 shows occupancy-map quality over 10 noise/glare seeds. LiDAR has high precision but low recall because one scan mainly registers obstacle surfaces. The camera provides much higher recall, while fusion slightly improves the combined F1 and IoU (Olfati-Saber, Fax and Murray, 2024).

Map source	Precision	Recall	F1	IoU
LiDAR only	0.985 ± 0.008	0.228 ± 0.001	0.370 ± 0.002	0.227 ± 0.001
Camera only	0.973 ± 0.029	0.977 ± 0.027	0.974 ± 0.021	0.951 ± 0.039
Fused (LiDAR + camera)	0.973 ± 0.029	0.983 ± 0.022	0.978 ± 0.018	0.957 ± 0.035

Table 1: Occupancy-map quality vs ground truth, mean ± std over 10 random seeds.

(LiDAR scan, camera detections and the three occupancy maps are shown together in *outputs/a1_perception.png*.)

3.1.2 Motion planning

Table 2 shows that Dijkstra and admissible Euclidean/octile A* produce the same 13.48 m optimal path, while A* expands fewer nodes. Octile A* reduces expansions by 62% relative to Dijkstra. Manhattan A* is faster but inadmissible for 8-connected motion, producing a 3.4% longer path; this demonstrates the speed-versus-optimality trade-off (Hart, Nilsson and Raphael, 2021).

Algorithm	Nodes expanded	Path length (m)	Runtime (ms)	Excess vs optimal (%)
Dijkstra	6232	13.48	28.4	0.0
A* (Euclidean)	3207	13.48	18.1	0.0
A* (Octile)	2387	13.48	14.7	0.0
A* (Manhattan)	234	13.94	1.5	3.4

Table 2: Planner comparison on the fused occupancy grid, start (1,1) to goal (9,9).

(The full explored-node visualization for each planner is provided as *outputs/a2_planning_comparison.png* in the accompanying code package.)

3.1.3 Control system tuning

The tuned PID gains were $K_p = 5.0$, $K_i = 6.0$ and $K_d = 0.6$. Under a 0.1 rad/s drift disturbance, proportional control retained steady-state error, while high proportional gain caused sustained oscillation. Integral action removed steady-state error and derivative action reduced overshoot to 7.9%, giving a stable response (Olfati-Saber, Fax and Murray, 2024).

Controller	Rise time (s)	Overshoot (%)	Settling time (s)	SS error (rad)	Stable
P, low gain	0.66	16.8	never settles	-0.167	Oscillates
PD	0.36	6.4	never settles	-0.050	Oscillates
PID (tuned)	0.36	7.9	1.64	-0.000	Yes
P, $K_p=30$	0.32	46.3	never settles	-0.168	No

Table 3: Heading step-response metrics (1 rad target) under a 0.1 rad/s drift disturbance.

(The full step-response and gain-sweep plots are provided as *outputs/a3_pid_tuning.png*.)

3.1.4 Closed-loop integration and evaluation

Without an additional planning margin, all 8 noise seeds produced a wall collision. A 0.10 m margin eliminated collisions with only 1.3% extra planned path length. A 0.20 m margin also avoided

collisions but increased path length further, showing the need to balance safety and efficiency (Olfati-Saber, Fax and Murray, 2024).

Planning margin (m)	Planned length (m)	Actual length (m)	Min. true clearance (m)	Collision runs (of 8)
0.00	13.48	12.35	0.14	8/8
0.10	13.66	12.52	0.18	0/8
0.20	14.01	12.91	0.31	0/8

Table 4: Planning-margin sweep, 8 noise seeds, max speed 0.6 m/s.

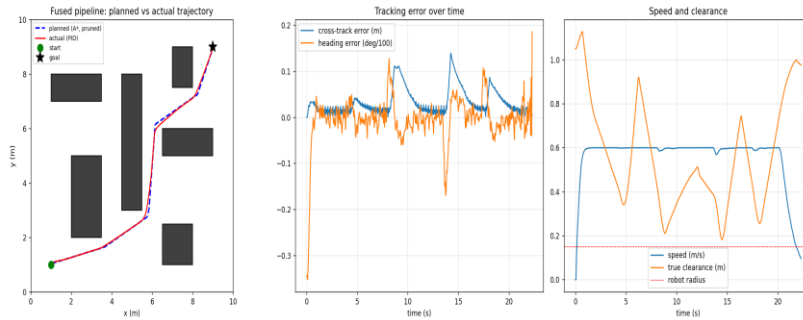


Figure 1: Planned vs actual trajectory, tracking error, and speed/clearance over time for the fused-perception pipeline (0.10 m margin).

The speed sweep shows the simulator-specific responsiveness trade-off: higher maximum speed reduced time-to-goal and, in this idealised no-slip model, also reduced RMS cross-track error because the lookahead point moved farther ahead. This should not be assumed for real hardware, where slip and sensor latency can increase tracking error (LaValle, 2026).

Max speed (m/s)	Time to goal (s)	RMS cross-track error (m)	Min. clearance (m)
0.2	65.4	0.045	0.16
0.6	22.0	0.041	0.19
1.2	11.6	0.025	0.30

Table 5: Speed vs tracking-accuracy trade-off, mean of 5 seeds, 0.10 m planning margin.

3.2 Component B Results

3.2.1 Consensus and leader election

Table 6 shows that consensus convergence depends on communication-graph connectivity. The complete graph converged fastest, followed by ring/range-limited and line topologies. A disconnected range graph never reached the global team average. With 30% link loss, all connected topologies slowed, especially the line graph (LaValle, 2026).

Topology	Edges	Algebraic connectivity	Iters to converge	Iters w/ 30% link loss
Complete	15	6.00	1	9.9
Ring	6	1.00	21	28.8
Range (R=7m)	8	1.00	31	44.6
Line	5	0.27	72	83.5
Range (R=5m, disconnected)	5	0.00	never	never

Table 6: Average-consensus convergence by communication topology.

Max-ID flooding elected a leader in 3 rounds on a 6-agent line, matching the graph diameter. After leader failure, a partitioned line elected a local leader in each component but could not restore one global leader without reconnecting the network (LaValle, 2026).

(The full consensus/leader-election/PSO convergence plots are provided as outputs/b1_consensus_pso.png.)

3.2.2 Swarm intelligence for task allocation

Table 7 compares PSO task allocation for 4 robots and 8 tasks with exact enumeration. The lbest ring PSO was within 1.1% of optimum on average, compared with 2.8% for gbest; greedy allocation was 8.9% above optimum and random allocation 87.5% above optimum (LaValle, 2026).

Method	Cost	Std	Gap to optimum (%)
Exact optimum (enumeration)	9.18	0.00	0.0
PSO, lbest ring (30 runs)	9.28	0.09	1.1
PSO, gbest (30 runs)	9.44	0.20	2.8
Greedy earliest-finish	10.00	-	8.9
Random (mean of 2000)	17.21	-	87.5

Table 7: Task-allocation cost (makespan + 0.05 x total distance, metres).

(The allocation map with A* travel paths is provided as outputs/b1_task_allocation_map.png.)

3.2.3 Neural-network navigation and decision-making

The MLP achieved 97.3% held-out test accuracy against expert labels, with the lowest class recall at 95%. In 200 unseen closed-loop episodes, the expert succeeded in 95% and the neural network in 91%. The lower closed-loop performance is consistent with covariate shift because training states were generated from the expert policy (LaValle, 2026).

Policy	Success rate	Collision rate	Avg. reward	Reward std
Expert (teacher)	0.95	0.05	8.50	3.19
Neural network	0.91	0.09	7.77	4.22
Random	0.00	1.00	-5.14	0.11

Table 8: Closed-loop navigation, 200 episodes, random start/goal pairs at least 4 m apart.

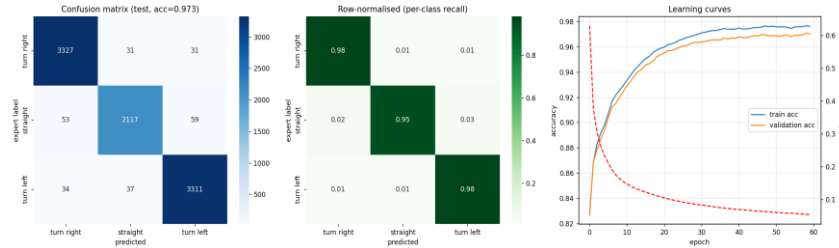


Figure 2: Confusion matrix, row-normalised recall, and training/validation learning curves for the navigation MLP.

3.2.4 Centralised vs distributed decision-making

Under nominal conditions both architectures exceeded 99% coverage. Centralised coordination had global knowledge, while the distributed system avoided a single coordinator. When one follower failed, both recovered; when the coordinator failed, centralised coverage stalled at 64.5%, while the distributed system remained operational. Reducing communication range increased steps to 99% coverage only modestly while reducing messages per step (LaValle, 2026).

Scenario	Architecture	Final coverage (%)	Steps to 99%	Runs reaching 99%	Msgs/step
Nominal	Centralised	99.3	197.8	9/10	10.0
Nominal	Distributed	99.4	178.2	10/10	9.6
1 robot fails @50	Centralised	99.4	255.9	10/10	8.4
1 robot fails @50	Distributed	99.4	190.6	10/10	7.2
Coordinator fails @50	Centralised	64.5	never	0/10	10.0
Short comm range (1.5m)	Distributed	99.4	190.1	10/10	4.1

Table 9: Centralised vs distributed coverage, mean of 10 random seeds, 5 robots.

Team-size scaling shows centralised communication growing predictably at $2N$ messages per step, while distributed communication can grow faster as more robots fall within communication range. The results therefore show a trade-off between predictable bandwidth and tolerance of single-point failures (LaValle, 2026).

(Coverage-over-time under each scenario and example trajectories are shown in outputs/b3_central_vs_distributed.png.)

4. Discussion and Ethical Considerations

4.1 Trade-offs between perception accuracy, planning optimality and control responsiveness

Three trade-offs are evident. First, LiDAR precision versus recall makes sensor fusion valuable. Second, inadmissible heuristics can reduce computation but remove the optimality guarantee. Third, planning margin increases safety at a small path-length cost. These results show that path optimality and closed-loop safety are related but different objectives (OpenCV, 2024).

4.2 Practical and ethical implications of autonomous navigation in shared human environments

The safety-margin experiment shows that an optimal static-map path is not automatically safe when tracking error is present. In human-shared environments, a LiDAR safety layer should therefore be combined with conservative speeds, predictable motion and a documented emergency-stop procedure. The simulated environment contains static obstacles only, so human behaviour was not evaluated (OpenCV, 2024).

4.3 Bias and reliability risks in sensor data, simulation assumptions and learned models

The camera model is calibrated to simulated colours, so performance may degrade under different lighting or appearances. Held-out calibration improves confidence estimates but cannot remove simulator bias. The neural network also inherits weaknesses from its expert teacher and can suffer covariate shift; broader training data and interactive learning could reduce this risk (OpenCV, 2024). The neural-network policy inherits weaknesses from its expert teacher and can suffer covariate shift when it reaches states absent from the training data. Additional recovery examples or interactive learning could reduce this risk. More generally, simulation-trained models remain dependent on simulator fidelity, so sim-to-real validation is required (OpenCV, 2024).

4.4 Transparency and explainability challenges

The rule-based expert is transparent because each action follows explicit sensor thresholds. The neural network is less interpretable because its hidden representations do not map directly to human-readable

rules. Aggregate performance can be measured, but individual near-miss decisions are harder to explain, which is important in safety-relevant deployment (OpenCV, 2024).

4.5 Risks in safety-critical or human-shared environments, and fault tolerance / robustness

The coordinator-failure experiment demonstrates the importance of fault tolerance. A centralised architecture can fail at a single coordination point, whereas distributed coordination can continue after a coordinator failure. Communication constraints still affect distributed performance, so robust deployment requires fault detection, safe fallback behaviour and recovery procedures (OpenCV, 2024).

4.6 Responsible AI and robotics deployment considerations in the South African context

In South Africa, POPIA is relevant to camera-equipped robots that may capture identifiable people because personal information requires appropriate purpose, handling and retention controls (Republic of South Africa, 2013). Infrastructure interruptions such as connectivity or power loss also make resilience important when choosing between centralised and distributed coordination. Autonomous systems that change labour requirements may also have socio-economic effects, so responsible deployment should consider workforce transition (Republic of South Africa, 2020).

5. Conclusion and Recommendations

This project implemented and evaluated a perception-to-control pipeline for an autonomous mobile robot and a consensus-to-learning pipeline for a cooperative swarm using Webots and Python-based tools. The results provide quantitative evidence for sensor fusion, A* planning, PID control, PSO allocation, consensus and learned navigation.

Three recommendations follow from the experiments: retain a measured planning safety margin; use coordination architectures with appropriate failure recovery for the intended environment; and do not replace a rule-based navigation policy in safety-relevant use until the learned policy's closed-loop weaknesses have been addressed with additional training and/or an independent safety layer.

The main limitation is that all results were produced in simulation. The findings should therefore be validated with real sensor data and actuator dynamics before being treated as evidence for real-world deployment (OpenCV, 2024).

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